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Copy-Move Forgery Detection in Digital Images

A Hybrid Approach Using SIFT and Custom Manual Algorithms

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GitHub Repository:

<https://github.com/omerkhann/CoMoFod>

Abstract— Copy-Move Forgery (CMF) is a prevalent class of digital image manipulation in which a region within an image is duplicated and transplanted to a different location in the same image, typically to conceal or fabricate visual content. Because the forged region shares the exact statistical characteristics of its source image—such as noise profile, illumination, and colour palette—conventional forgery-detection strategies routinely fail to expose it. This paper presents a robust Digital Image Processing (DIP) pipeline that combines the well-established Scale-Invariant Feature Transform (SIFT) for robust keypoint extraction with a suite of *manually implemented* mathematical algorithms. These include a vectorised Euclidean descriptor matcher, Lowe’s Ratio Test, a custom Direct Linear Transform (DLT) Homography solver, Sequential RANSAC clustering, and morphological binary-mask post-processing. Evaluation on 120 images drawn from the CoMoFoD benchmark yielded mean IoU scores of 0.32, 0.41, and 0.38 for translation, rotation, and scaling categories respectively. An interactive Gradio dashboard further enables real-time parameter exploration and forensic analysis, highlighting the acute sensitivity of feature-based detection to underlying image textures.

1. Introduction

The explosive democratisation of image-editing software has blurred the boundary between authentic photographs and fabricated visual narratives. Among the many categories of image forgery, *Copy-Move Forgery* (CMF) is particularly insidious: because the copied region originates from the same source photograph, its colour palette, noise profile, texture statistics, and illumination gradients are intrinsically consistent with the rest of the scene. Classical detection strategies that exploit noise inconsistencies, chromatic aberrations, or compression artefacts are therefore largely ineffective against CMF.

1.1 Motivation and Applications

The societal stakes of reliable CMF detection span several high-impact domains. In judicial proceedings, photographic evidence must be verifiably authentic, as undetected manipulation can directly subvert verdicts. In medical imaging, cloned tissues or manipulated X-rays can conceal tumors or fabricate diseases, risking severe diagnostic er-

rors. Furthermore, in academic research, the duplication of gel electrophoresis bands or microscopy data constitutes research misconduct. Reliable CMF detection acts as a critical countermeasure across all these disciplines.

1.2 Scope and Contributions

Unlike deep learning black-box methods (e.g., CNNs or GANs), which require massive computational overhead and labeled datasets, feature-based methods remain highly interpretable and mathematically transparent. This work makes the following contributions:

1. A fully manual, NumPy-based implementation of Euclidean descriptor matching, DLT Homography estimation, RANSAC inlier selection, and morphological post-processing.
2. A Sequential RANSAC strategy engineered to detect *multiple independent* cloned regions within a single image.
3. A quantitative evaluation on 120 CoMoFoD images spanning three geometric transformation categories (Translation, Rotation, Scaling).

2. Dataset

All experiments utilize the **CoMoFoD** (Copy-Move Forgery Database) benchmark, a rigorous standard in forensic imaging. To facilitate a balanced evaluation across different manipulation types, a curated 120-image subset was selected.

Table 1: Dataset composition.

ID Range	Transformation	Count
001–040	Translation only	40
041–080	Rotation applied	40
081–120	Scale change	40

Each entry in the dataset comprises the forged image containing the cloned region, and a highly precise binary ground-truth mask delineating the exact forged pixels, allowing for strict pixel-level evaluation metrics.

3. Methodology

3.1 Feature Extraction via SIFT

The pipeline uses the Scale-Invariant Feature Transform (SIFT) to locate and describe local features. SIFT operates by constructing a Difference-of-Gaussian (DoG) scale-space pyramid, identifying local extrema across scales to ensure scale invariance. Each valid keypoint is assigned a dominant orientation based on local image gradient directions, ensuring rotation invariance. Finally, a 128-dimensional orientation histogram descriptor is generated for each keypoint. These descriptors encapsulate the local texture, allowing the algorithm to match regions even if the cloned patch has been rotated or scaled.

3.2 Custom Descriptor Matching

Given descriptor matrices **A** and **B** extracted from the image, the complete $N \times M$

Euclidean distance matrix is computed. To maximize computational efficiency, this is executed *without loops* utilizing a vectorized algebraic expansion in NumPy.

Matches are filtered using **Lowe’s Ratio Test**. For a query descriptor, if the distance to the nearest match (d_1) and second-nearest match (d_2) yields a ratio $d_1/d_2 < \tau$, the match is retained. To prevent trivially overlapping patches in uniform areas from registering as forgeries, a **Spatial Distance Filter** enforces a minimum pixel-plane distance of 20 px between matched points.

3.3 Geometric Verification via RANSAC

Raw matches frequently contain false positives. To geometrically verify the matches, we utilize the Direct Linear Transform (DLT) to estimate a 3×3 homography matrix **H** from four correspondence pairs using Singular Value Decomposition (SVD).

Because an image may contain multiple distinct forged patches, standard RANSAC is insufficient. We implemented a **Sequential RANSAC** architecture: once a valid homography cluster is found and extracted, the algorithm repeats on the remaining outliers, iteratively discovering all independent forged regions until no valid clusters remain.

3.4 Morphological Post-Processing

Because keypoints are sparse, mathematically matching them generates a "dot cloud" rather than a solid shape. To compute Intersection over Union (IoU), we rasterize these points by drawing localized circles around the RANSAC inliers. We then apply custom 2D convolution logic to perform Morphological Closing (a dilation operation followed immediately by an erosion). This fills internal gaps and smooths the boundaries, yielding a dense binary mask comparable to the ground truth.

4. Evaluation and Results

4.1 Metrics

Let P and G denote the predicted and ground-truth binary masks respectively. The pipeline is evaluated using Intersection over Union (IoU) and the DICE Coefficient:

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}, \quad \text{DICE} = \frac{2|P \cap G|}{|P| + |G|} \quad (1)$$

4.2 Baseline Results

The evaluation was executed on the 120-image subset using a globally fixed parameter configuration: **Lowe’s Ratio** (τ) = **0.7**, and a **RANSAC threshold** (ϵ) = **5**.

Table 2 and Figure 1 summarize the mean performance.

Table 2: Baseline evaluation ($\tau = 0.7, \epsilon = 5$).

Category	Mean IoU	Mean DICE
Translation	0.3249	0.4904
Rotation	0.4116	0.5831
Scaling	0.3780	0.5486
Overall Average	0.3715	0.5407

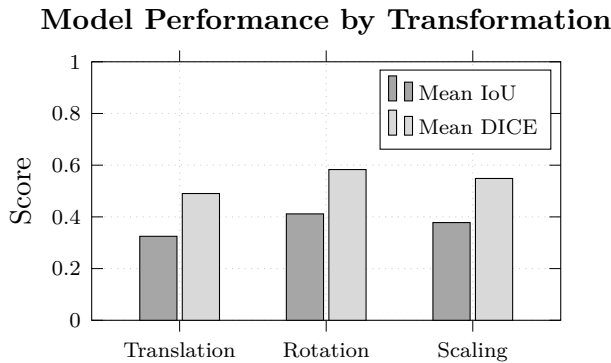


Figure 1: Mean detection metrics per category based on fixed global parameters.

4.3 Result Analysis

A granular review of the dataset reveals intense variance in detection success based

heavily on image content. Images with distinct, complex textures (such as 090_F.png and 006_F.png) achieved exceptional IoUs of 0.789 and 0.757 respectively. In these cases, SIFT extracts highly unique 128-D descriptors, allowing Lowe’s test to perfectly isolate the cloned region without false positives.

Conversely, a subset of images yielded an IoU of 0.0. This failure mode occurs predominantly in **Feature-Starved** or **Highly Repetitive** images (e.g., expansive skies, calm water, or flat painted walls). In feature-starved areas, SIFT fails to find enough DoG extrema to form a minimum 4-point RANSAC cluster. In repetitive textures (like a brick wall), the d_1 and d_2 nearest neighbors are functionally identical, causing Lowe’s Ratio test to reject valid matches.

5. Discussion and Limitations

The manual implementation of DLT and Sequential RANSAC provided complete mathematical auditability, confirming that closed-source API calls can be effectively replicated from first principles. Sequential RANSAC proved highly effective at identifying multiple disjoint clusters, which is essential for combating complex real-world forgeries.

However, the fixed-parameter evaluation highlights a fundamental limitation of feature-based methods: **Hyperparameter Sensitivity**. A Lowe’s ratio of 0.7 is a compromise. For heavily textured images, 0.7 is often too permissive and invites false positives. For flat, uniform textures, 0.7 is too strict and aggressively rejects true matches. This justifies the necessity of the interactive Gradio dashboard built alongside this project, which allows forensic analysts to iteratively adjust τ and ϵ to suit the specific noise profile of the image under investigation.

6. Conclusion

This project successfully implemented a transparent, mathematically rigorous pipeline for Copy-Move Forgery Detection. By eschewing black-box library calls for core geometric verification algorithms, the framework remains highly interpretable and pedagogically valuable. While baseline testing with fixed parameters ($\tau = 0.7$, $\epsilon = 5$) yielded a solid mean IoU of 0.37, the high variance across images confirms that a static, "one-size-fits-all" approach is insufficient for CMFD. Interactive parameter tuning, guided by an analyst's understanding of image texture, remains the most robust strategy for practical forensic analysis.